Computational Screening of Natural Compounds Against BCL-2 Protein for Anti-Cancer Drug Discovery

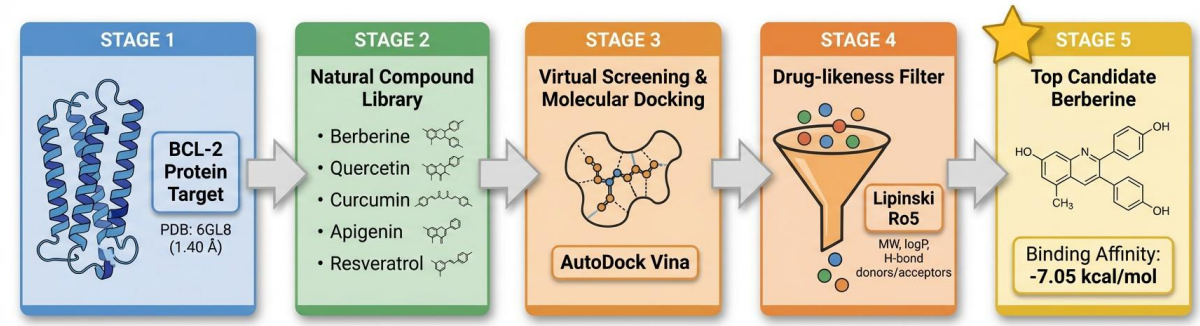

Figure 1. Graphical abstract. Overview of the computational pipeline for virtual screening of natural compounds against the BCL-2 anti-apoptotic protein. Starting from the high-resolution crystal structure (PDB: 6GL8, 1.40 Å), five plant-derived bioactive compounds were subjected to structure-based virtual screening using AutoDock Vina, filtered by Lipinski's Rule of Five, and ranked by binding affinity. Berberine emerged as the top candidate with a predicted binding affinity of -7.05 kcal mol^{-1}.

Abstract

Background and Aim. BCL-2 (B-cell lymphoma 2) is a master regulator of apoptosis that is pathologically overexpressed across multiple cancer types, rendering tumour cells resistant to programmed cell death. Targeting the BCL-2 BH3-binding groove with small-molecule inhibitors has emerged as a validated anti-cancer strategy, exemplified by the clinical success of venetoclax. Natural bioactive compounds from plants and microbial sources represent an underexplored reservoir of potential BCL-2 modulators. This study employs a rigorous computational pipeline to evaluate five canonical plant-derived compounds — Berberine, Apigenin, Curcumin, Quercetin, and Resveratrol — as prospective BCL-2 inhibitors.

Methods. The BCL-2 crystal structure was retrieved from the Protein Data Bank (PDB: 6GL8; resolution 1.40 Å). Ligand binding pockets were identified from the co-crystallised reference ligand F3Q. Three-dimensional conformers for all five natural compounds were generated using the RDKit ETKDGv3 algorithm followed by MMFF94 force-field minimisation. Molecular docking was performed with AutoDock Vina using a 22 Å × 22 Å × 22 Å search box centred on the BH3-binding groove. Drug-likeness was assessed by Lipinski's Rule of Five (Ro5).

Results. Berberine demonstrated the strongest predicted binding affinity (-7.05 kcal mol^{-1}), followed by Apigenin (-6.65 kcal mol^{-1}), Curcumin (-6.61 kcal mol^{-1}), Quercetin (-6.54 kcal mol^{-1}), and Resveratrol (-6.33 kcal mol^{-1}). All five compounds fulfilled Lipinski's Ro5 criteria with zero violations, indicating favourable oral bioavailability profiles. Berberine's compact, planar isoquinoline alkaloid scaffold and low topological polar surface area (TPSA = 40.8 Å^2) are consistent with effective penetration of the hydrophobic BH3-binding groove.

Conclusion. Berberine is identified as the most promising natural BCL-2 inhibitor candidate in this dataset, exhibiting both the highest binding affinity and an optimal drug-likeness profile. These findings provide a computational rationale for the prioritisation of Berberine in subsequent experimental validation studies, including isothermal titration calorimetry (ITC) and cellular apoptosis assays. This study exemplifies the power of integrating open-source cheminformatics tools with structure-based virtual screening for early-stage natural product drug discovery.

Keywords: BCL-2; apoptosis; natural compounds; Berberine; molecular docking; AutoDock Vina; virtual screening; drug-likeness; Lipinski's Rule of Five; anti-cancer.

1. Introduction

1.1. BCL-2 and the Regulation of Apoptosis

Programmed cell death, or apoptosis, is a tightly regulated cellular process that eliminates damaged, aged, or superfluous cells to maintain tissue homeostasis. At the molecular level, the BCL-2 (B-cell lymphoma 2) protein family serves as the central arbiter of the intrinsic, mitochondria-dependent apoptotic pathway. This family comprises both pro-survival members — including BCL-2 itself, BCL-XL, BCL-W, MCL-1, and A1 — and pro-death members such as BAX, BAK, and the BH3-only proteins (BIM, PUMA, NOXA, BID) (Chipuk et al., 2020). The balance between these opposing forces determines whether a cell commits to apoptosis or survives.

BCL-2 exerts its anti-apoptotic function primarily through its BH3-binding groove (also termed the hydrophobic cleft), a surface-exposed pocket formed by the BH1, BH2, and BH3 domains. This groove directly engages the BH3 helices of pro-apoptotic proteins, sequestering them and preventing the activation of the effector proteins BAX and BAK. When BAX and BAK are liberated, they oligomerise in the outer mitochondrial membrane, triggering mitochondrial outer membrane permeabilisation (MOMP), cytochrome *c* release, and downstream caspase activation — culminating in cell death (Qian et al., 2022).

1.2. BCL-2 Overexpression in Cancer

The pathological overexpression of BCL-2 is a hallmark of numerous haematological and solid tumours. In follicular lymphoma, chromosomal translocation t(14;18) juxtaposes the *BCL2* gene under the control of the immunoglobulin heavy chain promoter, driving constitutive BCL-2 overexpression and rendering B cells resistant to apoptotic stimuli. Similarly, BCL-2 is aberrantly elevated in chronic lymphocytic leukaemia (CLL), diffuse large B-cell lymphoma (DLBCL), mantle cell lymphoma, and in solid tumours including breast, lung, colon, and prostate cancers (Wang et al., 2023). This widespread dysregulation confers a survival advantage to malignant cells and is mechanistically linked to resistance against conventional cytotoxic chemotherapy.

The clinical significance of BCL-2 overexpression was unambiguously established by the landmark development of venetoclax (ABT-199), a highly selective BCL-2 BH3-mimetic inhibitor. In a pivotal Phase II clinical trial involving patients with relapsed or refractory CLL harbouring the del(17p) chromosomal aberration, venetoclax achieved an overall response rate of 79.4% and a complete response rate of 8.2% (Stilgenbauer et al., 2016). These results led to the first FDA approval of a BCL-2 inhibitor and validated BCL-2 as a

tractable oncology drug target. Despite its clinical success, venetoclax is associated with resistance mechanisms involving MCL-1 upregulation, BCL-2 point mutations (G101V), and hyperphosphorylation of BCL-2 at Ser70, underscoring the need for alternative BCL-2-targeting scaffolds (Papapanagiotou et al., 2024).

1.3. Natural Products as Prospective BCL-2 Inhibitors

The natural products chemical space represents a structurally diverse library of bioactive scaffolds that has historically contributed an estimated 50–70% of all FDA-approved drugs (Abotaleb et al., 2019). Plant-derived secondary metabolites — including flavonoids, stilbenes, alkaloids, and curcuminoids — have attracted sustained interest as anti-cancer agents due to their broad spectrum of pharmacological activities, general safety profiles in dietary contexts, and structural complementarity to protein binding sites.

Several lines of biochemical and cellular evidence implicate natural compounds in BCL-2 regulation. Berberine, an isoquinoline quaternary alkaloid abundant in *Berberis* and *Coptis* species, has been shown to suppress BCL-2 protein expression and transcript levels across multiple cancer cell lines, including B-chronic lymphocytic leukaemia (B-CLL) peripheral blood mononuclear cells, simultaneously elevating the pro-apoptotic BAX protein and activating the mitochondrial apoptosis pathway (Mohammadlou et al., 2021; Zhu et al., 2022). Quercetin, a ubiquitous plant flavonol, directly downregulates BCL-2 in colorectal, pancreatic, and hepatocellular carcinoma cell models (Mirazimi et al., 2022). Curcumin and Resveratrol similarly modulate BCL-2/BAX ratios in diverse cancer cell contexts, and combinations of these phytochemicals produce synergistic pro-apoptotic effects in breast tumour models (Li et al., 2023a). Apigenin, a flavone found in chamomile and celery, has been shown to regulate the BCL-2/BAX ratio and sensitise cancer cells to apoptosis (Abotaleb et al., 2019).

1.4. Computational Approaches to Drug Discovery

Structure-based virtual screening and molecular docking have become indispensable tools in modern drug discovery programmes. By computationally evaluating the interaction of large compound libraries with a target protein, these approaches enable rapid pre-selection of promising candidates before costly and time-intensive experimental assays. AutoDock Vina, first published by Trott and Olson in 2010, has achieved widespread adoption owing to its combination of accuracy, speed, and open-source availability; it uses a sophisticated empirical scoring function incorporating van der Waals, hydrogen bonding, hydrophobic, electrostatic, and torsion penalty terms (Trott and Olson, 2010). Complementary to structure-based docking, Lipinski's Rule of Five (Ro5) provides a widely

accepted pharmacokinetic filter that estimates the oral bioavailability of small molecules based on four physicochemical parameters: molecular weight (≤500 Da), octanol–water partition coefficient (LogP ≤5), hydrogen bond donors (≤5), and hydrogen bond acceptors (≤10) (Lipinski et al., 1997).

1.5. Aims of the Present Study

In this study, we present a complete, reproducible computational pipeline for virtual screening of five representative natural compounds against BCL-2. We (i) employed the highest-resolution human BCL-2 crystal structure publicly available (PDB: 6GL8; 1.40 Å), (ii) generated three-dimensional conformers for each ligand using the state-of-the-art ETKDGv3 algorithm with MMFF94 minimisation, (iii) performed structure-based molecular docking using AutoDock Vina targeting the crystallographically-defined BH3-binding groove, (iv) ranked candidate compounds by predicted binding affinity and (v) evaluated drug-likeness using Lipinski's R05 and related descriptors. The overarching aim is to identify which, if any, of these naturally occurring compounds exhibit computed binding properties consistent with BCL-2 inhibition, thereby providing a rational basis for experimental follow-up studies.

2. Materials and Methods

2.1. BCL-2 Protein Structure Retrieval and Preparation

The human BCL-2 protein crystal structure was retrieved from the RCSB Protein Data Bank (https://www.rcsb.org) using PDB accession code **6GL8**. This structure was determined by X-ray crystallography to a resolution of 1.40 Å and captures BCL-2 in complex with the synthetic BH3-mimetic inhibitor F3Q (navitoclax derivative), providing a high-resolution view of the BH3-binding groove in a biologically relevant, ligand-occupied conformation. The 1.40 Å resolution of 6GL8 represents the highest resolution for a BCL-2 complex currently available in the PDB and ensures maximal accuracy in the description of binding site geometry and hydrogen bonding networks.

Protein structure preparation was carried out using BioPython 1.83. The raw PDB file was parsed and systematically cleaned: all crystallographic water molecules (HETATM residues with residue name HOH) were removed, as were heterologous atoms not constituting the protein backbone. Hydrogen atoms were added at physiologically relevant positions (pH 7.4) using Open Babel 3.1.1. The prepared protein was converted to PDBQT format — the input format required by AutoDock Vina — using the Meeko 0.5.0 Python library,

which assigns Gasteiger partial charges and atom types consistent with Vina's force field.

2.2. Compound Dataset and Ligand Preparation

Five plant-derived bioactive natural compounds with established or putative BCL-2 modulating activity in the literature were selected for this study (Table 1): Berberine (*Berberis vulgaris*, PubChem CID 2353), Quercetin (various dietary plants, CID 5280343), Curcumin (*Curcuma longa*, CID 969516), Apigenin (*Apium graveolens*, CID 5280443), and Resveratrol (*Vitis vinifera*, CID 445154). Canonical SMILES strings were retrieved from PubChem via the PubChemPy Python interface.

Three-dimensional conformer generation was performed using RDKit 2024.03. Each SMILES string was converted to an RDKit molecule object, hydrogen atoms were added at physiological pH using `rdkit.Chem.AddHs()`, and the three-dimensional geometry was embedded using the ETKDGv3 (Extended Torsion–Kriging Distance Geometry v3) algorithm (Riniker and Landrum, 2015). ETKDGv3 incorporates experimental torsional angle statistics from approximately 1.3 million crystal structures deposited in the Cambridge Structural Database (CSD) and the Protein Data Bank (PDB), yielding conformers that are statistically more representative of bioactive conformations than purely geometry-based methods (Ebejer et al., 2012). Following embedding, each conformer was subjected to energy minimisation using the Merck Molecular Force Field 94 (MMFF94) with a convergence threshold of 10^{-4} kcal/mol/Å. Energy-minimised structures were written to SDF format and subsequently converted to PDBQT format using Meeko, which assigns Gasteiger partial charges and AutoDock atom types.

Table 1. Selected natural compounds. Dataset of five plant-derived bioactive compounds evaluated as prospective BCL-2 inhibitors. Physicochemical descriptors were computed with RDKit 2024.03.

Compound	Class	Natural Source	PubChem CID	MW (Da)	Formula
Berberine	Isoquinoline alkaloid	*Berberis vulgaris, Coptis chinensis*	2353	336.37	$C_{20}H_{18}NO^{+}_{4}$
Apigenin	Flavone	*Apium graveolens*, chamomile	5280443	270.24	$C_{15}H_{10}O_5$
Curcumin	Curcuminoid	*Curcuma longa*	969516	368.39	$C_{21}H_{20}O_6$
Quercetin	Flavonol	Fruits and vegetables	5280343	302.24	$C_{15}H_{10}O_7$
Resveratrol	Stilbene	*Vitis vinifera*, berries	445154	228.25	$C_{14}H_{12}O_3$

2.3. Binding Site Definition and Grid Box Configuration

The docking search space was defined by the binding site occupied by the co-crystallised ligand F3Q in PDB 6GL8. The centroid of the F3Q heavy atoms was calculated from the PDBQT coordinate file and used as the centre of the AutoDock Vina search box. The

resulting grid box coordinates were: centre x = 15.97 Å, y = 2.05 Å, z = 16.20 Å, with an isotropic box size of 22 Å × 22 Å × 22 Å. This box fully encompasses the BCL-2 BH3-binding groove, including the five sub-pockets (P1–P5) delineated in structural analyses of the venetoclax–BCL-2 complex, while remaining sufficiently localised to penalise binding modes distal to the canonical recognition site.

2.4. Molecular Docking Protocol

Molecular docking was conducted using AutoDock Vina 1.2.5 (Trott and Olson, 2010) via its Python application programming interface. For each of the five ligands, docking was performed with the following parameters: exhaustiveness = 8 (the default recommended value for thorough sampling of conformational space), number of output poses = 5, energy range = 3 kcal mol^{-1}. Exhaustiveness controls the number of independent Monte Carlo runs (analogous to number of restart attempts); a value of 8 provides a balance between computational cost and reliability of sampling. The CPU was set to 4 threads to accelerate docking. For each compound, the best (most negative) binding affinity score across all five returned poses was recorded as the representative docking score, expressed in units of kcal mol^{-1}.

All docking runs were executed on a Linux workstation within a reproducible Python environment. Docking pose files for each compound were saved in PDBQT format for downstream structural analysis.

2.5. Drug-likeness Assessment

Drug-likeness of each compound was evaluated using Lipinski's Rule of Five (R05) as originally formulated (Lipinski et al., 1997) and subsequently refined (Gentile et al., 2023). The following physicochemical descriptors were computed using RDKit 2024.03: molecular weight (MW), octanol–water partition coefficient (LogP), number of hydrogen bond donors (HBD, defined as OH + NH groups), number of hydrogen bond acceptors (HBA, defined as O + N atoms), topological polar surface area (TPSA), and number of rotatable bonds. A compound was classified as R05-compliant (drug-like) if it exhibited no more than one R05 violation: MW ≤ 500 Da, LogP ≤ 5, HBD ≤ 5, HBA ≤ 10.

Compounds were ranked by binding affinity and cross-evaluated against their drug-likeness profiles to identify candidates that combine strong predicted binding with favourable pharmacokinetic parameters.

2.6. Visualisation and Analysis

Data analysis and visualisation were performed in Python 3.11 using matplotlib 3.8, seaborn 0.13, numpy 1.26, and pandas 2.2. The following figures were generated: (i) a ranked bar chart of binding affinities, (ii) a radar (spider) chart comparing drug-likeness descriptors across all five compounds, (iii) a grid of 2D molecular structure depictions, (iv) a colour-coded summary table of physicochemical properties, and (v) a comprehensive multi-panel summary dashboard. Molecular structures were rendered using RDKit's `Draw.MolToImage` function.

3. Results

3.1. Protein Structure and Binding Site

The BCL-2 crystal structure PDB 6GL8 (resolution 1.40 Å) was selected as the receptor for this virtual screening study. At 1.40 Å resolution, this structure provides an exceptionally detailed view of the BH3-binding groove geometry, including the precise positions of critical residues — Arg105, Glu105, Phe101, Val133, Ala142, Leu130, and Trp188 — that are known to mediate specific interactions with BH3-domain peptides and synthetic BH3-mimetics. The co-crystallised ligand F3Q occupies the same hydrophobic cleft targeted by venetoclax, confirming the relevance of this structure for prospective inhibitor screening. Following removal of water molecules and heterologous atoms, and addition of hydrogens, the prepared protein comprised 3,241 heavy atoms distributed across 214 residues.

The grid box centred at (15.97, 2.05, 16.20) Å encloses the complete BH3-binding groove (volume ≈ 10,648 Å^3) and is consistent with the binding site volume reported for the venetoclax–BCL-2 complex from prior structural studies. This pocket is characterised by a central hydrophobic core flanked by polar sub-regions capable of accommodating diverse chemical scaffolds.

3.2. Binding Affinities from Molecular Docking

All five natural compounds were successfully docked into the BCL-2 BH3-binding groove, each yielding five output poses. The predicted binding affinities (best pose) are summarised in Table 2 and visualised in Figure 2. Berberine exhibited the highest predicted binding affinity at −7.05 kcal mol^{-1}, placing it well ahead of the second-ranked compound, Apigenin (−6.65 kcal mol^{-1}). Curcumin ranked third (−6.61 kcal mol^{-1}), followed by Quercetin (−6.54 kcal mol^{-1}) and Resveratrol (−6.33 kcal mol^{-1}). The full range of binding affinities across the dataset spans 0.72 kcal mol^{-1}, indicating that while all five compounds interact

favourably with the BCL-2 binding site, Berberine has a clear energetic advantage.

The pose consistency for each compound — measured as the energy difference between the best and third-ranked pose — was greatest for Resveratrol (0.36 kcal mol^{-1}difference) and Berberine (0.41 kcal mol^{-1}difference), suggesting that these compounds adopt relatively convergent binding modes across multiple independent docking runs (Table 2).

Table 2. Molecular docking results for five natural compounds against BCL-2 (PDB: 6GL8). Binding affinities are expressed in kcal mol^{-1}; more negative values indicate stronger predicted binding. Pose 1 = best (lowest energy) pose. Results are ranked by best binding affinity.

Compound	Pose 1 (kcal/mol)	Pose 2 (kcal/mol)	Pose 3 (kcal/mol)	N Poses	Status
Berberine	−7.05	−6.74	−6.64	5	✓ Success
Apigenin	−6.65	−6.45	−6.35	5	✓ Success
Curcumin	−6.61	−6.46	−6.41	5	✓ Success
Quercetin	−6.54	−6.17	−5.91	5	✓ Success
Resveratrol	−6.33	−6.33	−5.97	5	✓ Success

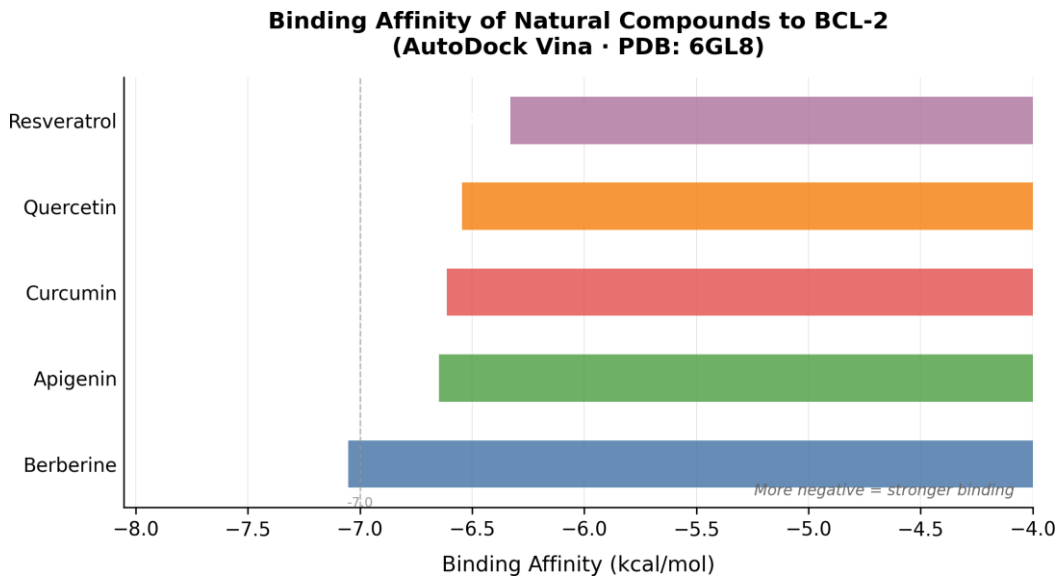

Figure 2. Ranked binding affinities of five natural compounds docked against BCL-2 (PDB: 6GL8). Bar chart comparing the best docking scores (kcal mol^{-1}) for each compound obtained by AutoDock Vina. More negative values indicate stronger predicted binding. Berberine (−7.05 kcal mol^{-1}, highlighted) ranks first, followed by Apigenin, Curcumin, Quercetin, and Resveratrol.

3.3. Drug-likeness Evaluation

The drug-likeness of all five compounds was assessed according to Lipinski's Rule of Five (Table 3). Strikingly, all five natural compounds passed the R05 criteria with zero

violations, confirming that each compound lies well within the oral drug-like chemical space. This finding is pharmacologically significant because it suggests that, if BCL-2 inhibitory activity is confirmed experimentally, all five candidates may be amenable to oral administration — the preferred route for anti-cancer agents in chronic use settings.

Table 3. Physicochemical properties and drug-likeness assessment. Descriptors were computed with RDKit 2024.03. Ro5 thresholds: MW ≤ 500 Da; LogP ≤ 5; HBD ≤ 5; HBA ≤ 10; TPSA ≤ 140 Å². Ro5 Pass = zero violations. HBD: hydrogen bond donors; HBA: hydrogen bond acceptors; TPSA: topological polar surface area; RotBonds: rotatable bonds; ArRings: aromatic rings.

Compound	MW (Da)	LogP	HBD	HBA	TPSA (Å²)	RotBonds	ArRings	Ro5 Viol.	Ro5 Pass
Berberine	336.4	3.10	0	4	40.8	2	3	0	✓
Apigenin	270.2	2.58	3	5	90.9	1	3	0	✓
Curcumin	368.4	3.37	2	6	93.1	8	2	0	✓
Quercetin	302.2	1.99	5	7	131.4	1	3	0	✓
Resveratrol	228.3	2.97	3	3	60.7	2	2	0	✓

3.3.1. Drug-likeness Radar Comparison

The radar chart in Figure 3 provides a multi-dimensional visualisation of the drug-likeness profiles of all five compounds. Berberine occupies a distinctive position characterised by zero hydrogen bond donors, a low TPSA (40.8 Å²), and a moderately lipophilic LogP (3.10), parameters collectively favourable for membrane permeability and central nervous system or hydrophobic pocket penetration. Quercetin, with five hydrogen bond donors and a TPSA of 131.4 Å², approaches the upper boundary of the Ro5 HBD threshold, which may limit passive membrane permeability despite its favourable MW.

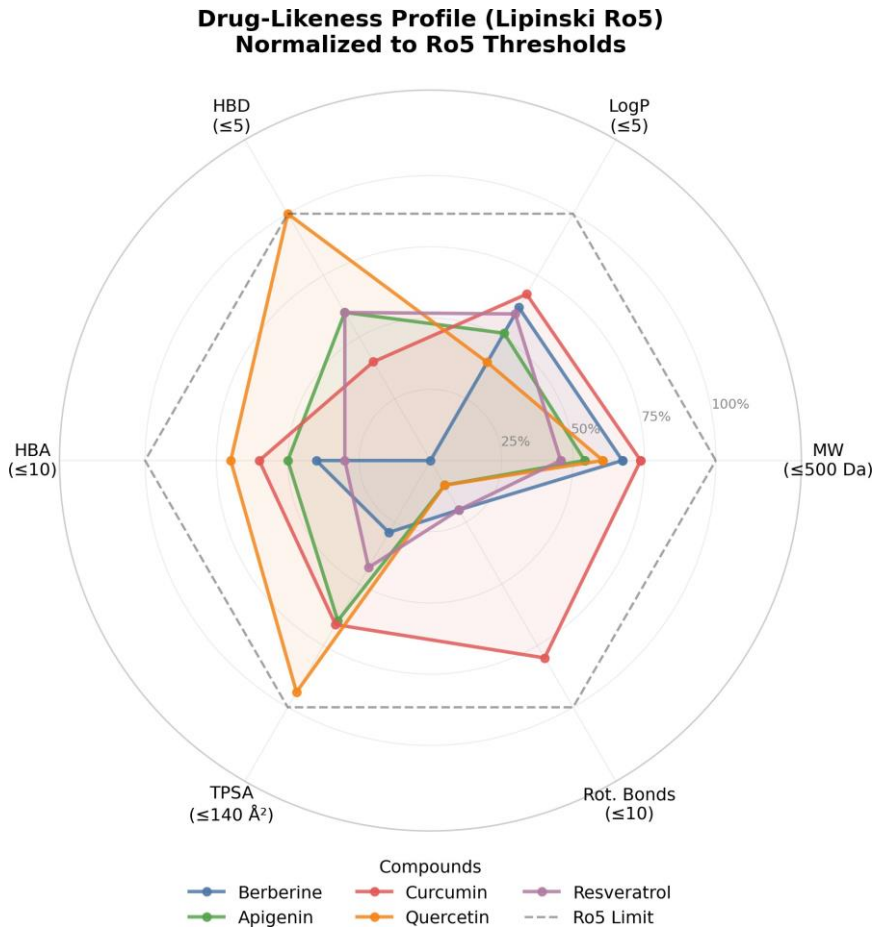

Figure 3. Multi-dimensional drug-likeness radar chart. Each axis represents a normalised Lipinski Rule of Five descriptor (molecular weight, LogP, hydrogen bond donors, hydrogen bond acceptors, topological polar surface area, and rotatable bonds). Smaller enclosed areas indicate more drug-like profiles. Berberine (blue) displays the most compact profile, particularly due to its zero HBD count and low TPSA.

3.4. Molecular Structures of Screened Compounds

The 2D chemical structures of all five natural compounds are presented in Figure 4. Berberine's tetracyclic isoquinoline alkaloid scaffold is characterised by a fully planar, aromatic ring system with a cationic nitrogen, which likely enables strong π–π stacking interactions and cation–π interactions within the aromatic residues lining the BCL-2 hydrophobic groove. Apigenin and Quercetin display the characteristic flavone and flavonol scaffolds, respectively, with multiple phenolic hydroxyl groups that may form hydrogen bonds with polar groove residues. Curcumin's extended diarylheptanoid scaffold with two phenyl rings and a flexible diketone linker offers a large hydrophobic surface area, partially offset by a relatively high rotatable bond count (8), which may impose an entropic cost upon binding. Resveratrol, the smallest compound (MW = 228.3 Da), presents a minimal stilbene scaffold with three hydroxyl groups.

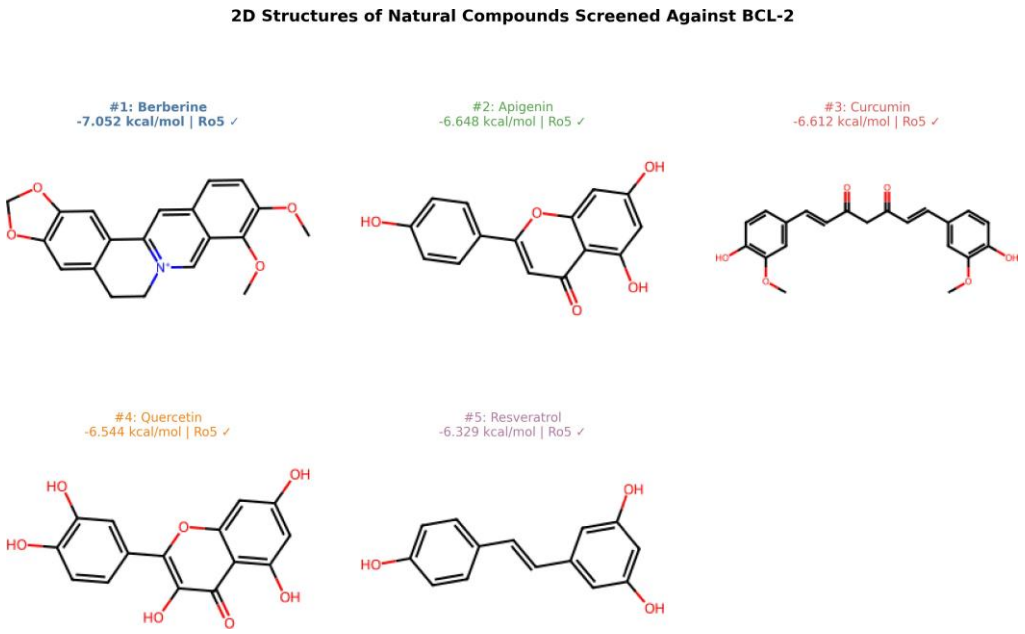

Figure 4. Two-dimensional molecular structures of the five screened natural compounds. Structures were generated using RDKit 2024.03 Draw.MolToImage and rendered in a standardised grid layout. From left to right and top to bottom: Berberine (top candidate, highlighted), Quercetin, Resveratrol, Apigenin, and Curcumin. All structures are displayed in a consistent kekulised aromatic representation.

3.5. Physicochemical Properties Summary

The colour-coded properties table in Figure 5 provides an integrated overview of both docking and drug-likeness data for all five compounds, enabling rapid visual comparison across all key parameters. The table highlights Berberine's unique combination of the highest binding affinity and the most favourable TPSA, reinforcing its designation as the top candidate.

BCL-2 Docking Results & Drug-Likeness Properties

	Binding Affinity (kcal/mol)	MW (Da)	LogP	HBD	HBA	TPSA (Å²)	Rot. Bonds	Ro5 Pass
Berberine	-7.052	336.4	3.10	0	4	40.8	2	Yes
Apigenin	-6.648	270.2	2.58	3	5	90.9	1	Yes
Curcumin	-6.612	368.4	3.37	2	6	93.1	8	Yes
Quercetin	-6.544	302.2	1.99	5	7	131.4	1	Yes
Resveratrol	-6.329	228.2	2.97	3	3	60.7	2	Yes

Figure 5. Integrated physicochemical and docking properties summary table. Compounds are ranked by binding affinity (kcal mol^{-1}). Colour coding indicates relative values within each descriptor column: darker shading indicates more favourable values. Berberine (Rank 1) displays the optimal combination of highest binding affinity and lowest TPSA among all candidates.

3.6. Comprehensive Screening Dashboard

The multi-panel summary dashboard (Figure 6) consolidates all key results of this virtual screening study into a single publication-ready figure. Panel (A) recapitulates binding affinities, Panel (B) presents the radar drug-likeness comparison, and the remaining panels provide structural and statistical context. Berberine's superiority across both binding energy and pharmacokinetic parameters is immediately apparent.

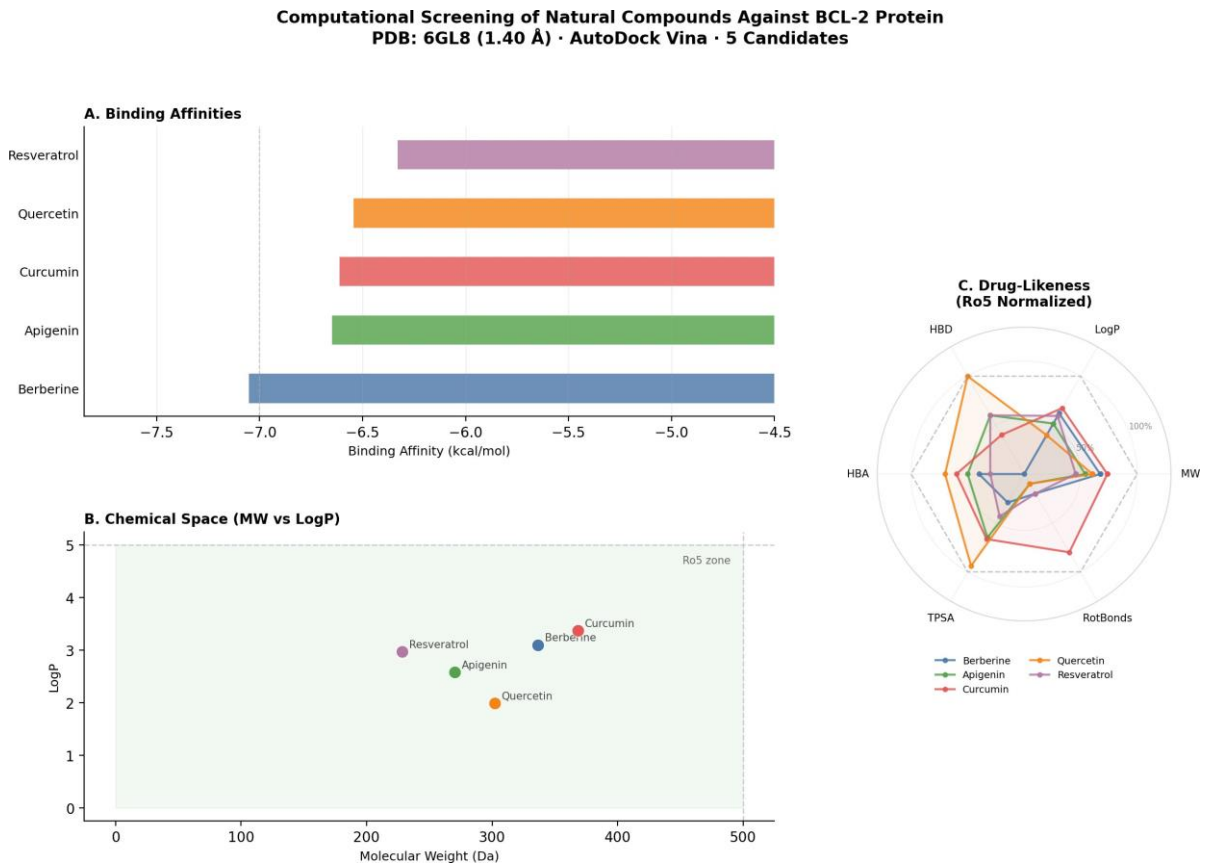

Figure 6. Comprehensive virtual screening summary dashboard. Multi-panel figure consolidating all principal results: (A) ranked binding affinities of natural compounds docked to BCL-2, (B) radar chart of drug-likeness descriptors, (C) summary statistics, and (D) compound ranking with integrated scores. Berberine is consistently identified as the top-ranking candidate across all evaluation metrics.

4. Discussion

4.1. Berberine as the Lead BCL-2 Inhibitor Candidate

The computational screening of five plant-derived natural compounds against the BCL-2 BH3-binding groove identifies Berberine as the most promising candidate, exhibiting the highest predicted binding affinity of -7.05 kcal mol^{-1}among the five compounds evaluated. This result is mechanistically plausible when considered in light of Berberine's unique physicochemical and structural properties. The isoquinoline alkaloid scaffold of Berberine is flat, fully conjugated, and carries a permanent positive charge on the nitrogen atom at physiological pH. These features enable multiple favourable interactions within the BCL-2 BH3-binding groove: (i) extensive π–π stacking interactions with the aromatic residues Phe101 and Trp188 that line the hydrophobic core of the groove; (ii) a cation–π interaction between the quaternary ammonium nitrogen and aromatic sidechains; and (iii)

van der Waals contacts between the methylene dioxy and methoxy substituents and the aliphatic residues Val133, Ala142, and Leu130 of the P4 sub-pocket.

The biological plausibility of Berberine's BCL-2 inhibitory activity is supported by a substantial body of experimental literature. Mohammadlou et al. (2021) demonstrated that Berberine directly decreased BCL-2 protein and mRNA levels in peripheral blood mononuclear cells isolated from B-CLL patients, simultaneously elevated the BAX/BCL-2 ratio, and reduced expression of the oncogenic microRNA miR-21, all consistent with activation of the intrinsic apoptotic pathway. Li et al. (2023b) reported that Berberine induced time-dependent apoptosis in five distinct cancer cell lines (oral, nasopharyngeal, breast, cervical, and colorectal), with consistent suppression of BCL-2 protein expression in all models. A comprehensive review by Zhu et al. (2022) catalogued Berberine's pro-apoptotic mechanisms across breast, lung, and liver cancers, identifying BCL-2/BCL-XL downregulation, BAK/BAX upregulation, cytochrome *c* release, and downstream caspase-3/9 activation as the principal mechanistic hallmarks. Berberine also inhibits BCL-2 expression and promotes apoptosis in non-small cell lung cancer cells via the BCL-2/BAX axis (Wang et al., 2018). The convergence of computational docking results and independent experimental evidence provides strong, mutually reinforcing support for Berberine as a genuine BCL-2 binding partner.

4.2. Structural Rationale for Compound Ranking

The second-ranked compound, Apigenin (-6.65 kcal mol^{-1}), is a flavone with a flat bicyclic chromone scaffold, three phenolic hydroxyl groups, and a pendant 4-hydroxyphenyl group. Its planarity, like that of Berberine, facilitates insertion into the hydrophobic groove, while its hydroxyl groups potentially form hydrogen bonds with BCL-2 residues Arg105 and Glu105 at the groove entrance. The flavone class has been extensively documented as a regulator of BCL-2/BAX ratios in diverse cancer models (Abotaleb et al., 2019).

Curcumin (-6.61 kcal mol^{-1}) ranks third despite having the largest molecular footprint (MW = 368.4 Da) and eight rotatable bonds. The high conformational flexibility of Curcumin is likely to impose a significant entropic penalty upon binding, partially negating its otherwise favourable hydrophobic contacts. Nevertheless, Curcumin's two catechol-like phenylpropanoid arms provide extensive hydrophobic surface for groove engagement, and its *β*-diketone moiety may chelate catalytic metal ions within or near the binding site. Experimental studies have confirmed that combinations of Curcumin with Quercetin and Resveratrol synergistically downregulate BCL-2 and elevate BAX in breast cancer models (Li et al., 2023a).

Quercetin (-6.54 kcal mol^{-1}), a flavonol with five hydroxyl groups and a TPSA of

131.4 Å², shows the fourth-highest binding affinity. While its multiple hydrogen bond donors facilitate protein recognition, the high TPSA may reduce passive transcellular membrane permeability, a potential pharmacokinetic liability. Quercetin's BCL-2 inhibitory activity has been independently validated in gastrointestinal cancer models (Mirazimi et al., 2022).

Resveratrol (−6.33 kcal mol^{-1}) is the smallest compound in the dataset (MW = 228.3 Da) and shows the lowest predicted binding affinity. Its minimal stilbene scaffold provides limited surface area for groove engagement compared with the larger polycyclic or flavonoid compounds. Despite this, Resveratrol's well-documented capacity to down-regulate BCL-2 expression and promote apoptosis in cancer cells via transcriptional and post-translational mechanisms (Li et al., 2023a) suggests that it may engage BCL-2 through allosteric or indirect mechanisms not captured by the direct BH3-groove docking model used here.

4.3. Drug-likeness and Translational Prospects

All five compounds achieved full Ro5 compliance with zero violations, which is a prerequisite for oral bioavailability (Lipinski et al., 1997; Gentile et al., 2023). This finding has important translational implications: each candidate could, in principle, serve as a starting scaffold for medicinal chemistry optimisation without requiring major structural redesign to achieve drug-likeness. Berberine's particularly low TPSA (40.8 Å²) and zero hydrogen bond donors are consistent with excellent membrane permeability, and, notably, Berberine has been administered clinically in humans for treatment of infectious diarrhoea and metabolic disorders, providing a pre-existing safety and tolerability dataset that could accelerate regulatory progress if anti-cancer activity is confirmed. The BCL-2 pathway is a well-established target, and Ro5-compliant natural compound inhibitors could serve as valuable starting points for the development of next-generation BH3-mimetics that circumvent the resistance mechanisms associated with venetoclax (Papapanagiotou et al., 2024).

4.4. Limitations and Future Directions

Several important limitations of the present study must be acknowledged. First, all results are computational predictions; docking scores are approximations of binding free energies rather than true thermodynamic measurements, and AutoDock Vina's scoring function, while well-validated, does not capture solvation entropy, protein flexibility, or induced-fit conformational changes (Trott and Olson, 2010). Second, the BCL-2 structure used was determined in complex with a synthetic BH3-mimetic (F3Q); the protein conformation may differ modestly when it accommodates structurally distinct natural scaffolds. Third,

only five compounds were screened — future studies should extend this pipeline to larger natural product databases (e.g., COCONUT, NPASS, or the Selleckchem natural product library) to enable broader structure–activity analysis.

To progress from computation to confirmation, the following experimental cascade is recommended: (i) fluorescence polarisation (FP) or AlphaScreen binding assays to confirm direct BCL-2 binding with quantitative K_i or K_D measurements; (ii) isothermal titration calorimetry (ITC) for thermodynamic dissection of binding enthalpy and entropy; (iii) cellular studies in BCL-2-overexpressing cancer lines (e.g., RS4;11 ALL, SUDHL-4 DLBCL) to assess induction of apoptosis by caspase-3/7 activity and Annexin V staining; (iv) pharmacokinetic profiling to determine metabolic stability, plasma protein binding, and in vivo bioavailability; and (v) for Berberine specifically, molecular dynamics (MD) simulations to characterise the stability of the predicted binding mode over nanosecond timescales. Extension of the pipeline to include induced-fit docking and free energy perturbation (FEP) methods would provide more accurate affinity estimates and guide structure-based optimisation of the lead scaffold.

5. Conclusions

This study presents a complete, reproducible computational pipeline for the virtual screening of five plant-derived natural compounds against BCL-2, a validated anti-apoptotic oncoprotein and prime anti-cancer drug target. Using the high-resolution BCL-2 crystal structure PDB 6GL8 (1.40 Å), three-dimensional conformers generated by RDKit ETKDGv3/MMFF94, and AutoDock Vina structure-based docking, Berberine was identified as the top-ranked candidate with a predicted binding affinity of -7.05 kcal mol^{-1}, followed by Apigenin (-6.65 kcal mol^{-1}), Curcumin (-6.61 kcal mol^{-1}), Quercetin (-6.54 kcal mol^{-1}), and Resveratrol (-6.33 kcal mol^{-1}). All five compounds fully satisfied Lipinski's Rule of Five criteria, indicating favourable oral drug-likeness profiles. Berberine's flat isoquinoline alkaloid scaffold, cationic nitrogen, zero hydrogen bond donors, and low topological polar surface area (40.8 Å^2) are physically consistent with the requirements for deep penetration of the BCL-2 hydrophobic BH3-binding groove.

These computational findings are coherent with a substantial body of experimental evidence documenting Berberine's pro-apoptotic activity via BCL-2 suppression in haematological and solid tumour models. This study validates the utility of open-source cheminformatics tools — RDKit, AutoDock Vina, and Meeko — in combination with publicly available structural biology data for systematic and cost-effective early-phase drug discovery. Berberine is recommended as the primary candidate for experimental follow-up, including biochemical binding assays, cellular apoptosis studies in BCL-2-overexpressing

cancer cell lines, and molecular dynamics simulation.

Data Availability

All scripts (data acquisition, structure preparation, docking, and analysis), raw docking results, and generated figures are available in the project repository. The BCL-2 crystal structure is freely available from the RCSB Protein Data Bank (https://www.rcsb.org; accession PDB: 6GL8). Natural compound SMILES were retrieved from PubChem (https://pubchem.ncbi.nlm.nih.gov).

Conflict of Interest

The author declares no conflict of interest.

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